

Lab 2, Act 2 – Answer Key

Could a different function fit the tadpole data better?

NRES 746 – Instructor Answer Key (not for student distribution)

Part 1: Two new candidates

1a. Setting $c = 1$ in $y = a x^c e^{-cx/p}$ gives $y = a x e^{-x/p}$. Comparing this to the plain Ricker, $y = a x e^{-bx}$, the exponent terms match when $b = 1/p$ (equivalently $p = 1/b$) – so the flexible Ricker is a real generalization: setting $c = 1$ recovers the original Ricker, with p playing the role of “peak location” wherever $1/b$ used to.

1b. Differentiating $y = a x^c e^{-cx/p}$ with the product rule (treating $u = x^c$, $v = e^{-cx/p}$) and the chain rule for v :

$$\frac{dy}{dx} = a \left[c x^{c-1} e^{-cx/p} + x^c e^{-cx/p} \left(-\frac{c}{p} \right) \right] = a c x^{c-1} e^{-cx/p} \left(1 - \frac{x}{p} \right)$$

This is zero when $x = 0$ (a trivial/edge case) or when $1 - x/p = 0$, i.e. $x = p$. So the critical point is at $x = p$, confirming p is the peak location by construction.

Candidate 2 setup (Holling type IV). $x^* = -2b/c$ for $b = 115.09$, $c = -21.49$:

```
b_h <- 115.09; c_h <- -21.49
x_star <- -2 * b_h / c_h
x_star
```

```
## [1] 10.71103
```

1c, second part. As $x \rightarrow \infty$: the numerator’s leading term is ax^2 and the denominator’s leading term is also x^2 , so $y \rightarrow ax^2/x^2 = a$ – a finite, positive asymptote, *not* zero. This is different from the Ricker family, which decays all the way to zero as $x \rightarrow \infty$. (This is the point of the “heads-up” box in the handout: Bolker’s parameterization of Holling type IV keeps a nonzero floor, unlike the more commonly cited Andrews/Monod-Haldane version.)

1d. Correct R functions:

```
flex_ricker <- function(x, a, p, c) a * x^c * exp(-c * x / p)
holling4 <- function(x, a, b, c) a * x^2 / (b + c * x + x^2)
```

Group prediction. Open-ended – there’s no “correct” prediction here, since the point of 2d is that intuition (more parameters = better) is a trap. A thoughtful answer might reasonably guess Holling IV or flexible Ricker will look better since they have an extra parameter; the payoff comes from testing that prediction against what “better” actually means (lowest SSR vs. best-looking curve vs. most trustworthy model), not from getting it “right” in advance.

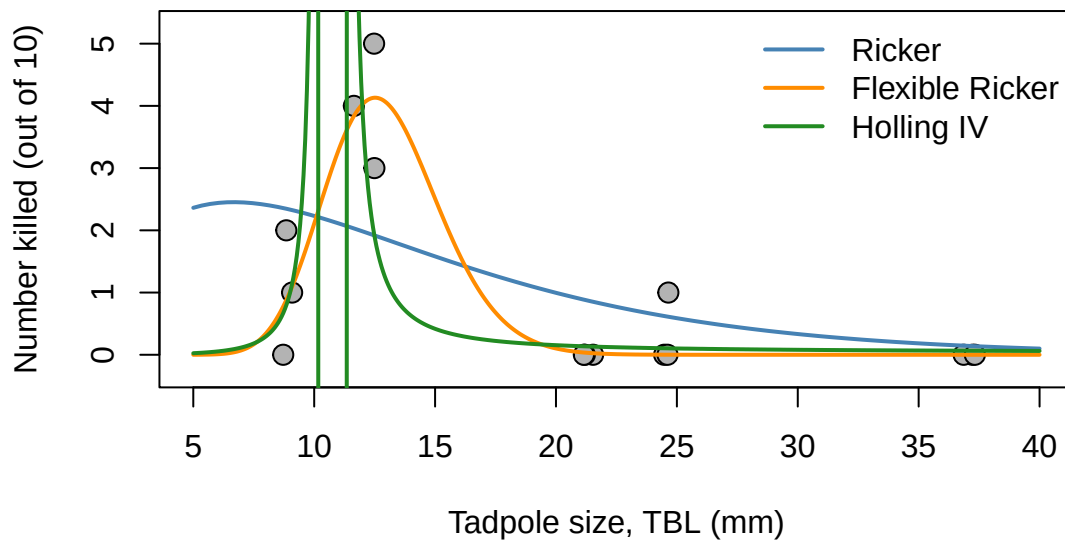
Part 2: Implement and compare

```
a_rk <- 1; b_rk <- 0.15
a_fr <- 3.68e-19; p_fr <- 12.52; c_fr <- 28.72
a_h4 <- 0.0329; b_h4 <- 115.09; c_h4 <- -21.49
```

```

set.seed(1)
x_j <- frogs$TBL + runif(nrow(frogs), -0.6, 0.6)
plot(x_j, frogs$Kill, xlim = c(5, 40), ylim = c(-0.3, 5.3),
     xlab = "Tadpole size, TBL (mm)", ylab = "Number killed (out of 10)",
     pch = 21, bg = "grey70", cex = 1.4)
curve(ricker(x, a_rk, b_rk), from = 5, to = 40, lwd = 2, col = "steelblue", add = TRUE)
curve(flex_ricker(x, a_fr, p_fr, c_fr), from = 5, to = 40, lwd = 2, col = "darkorange",
      add = TRUE, n = 500)
curve(holling4(x, a_h4, b_h4, c_h4), from = 5, to = 40, lwd = 2, col = "forestgreen",
      add = TRUE, n = 500)
legend("topright", legend = c("Ricker", "Flexible Ricker", "Holling IV"),
      col = c("steelblue", "darkorange", "forestgreen"), lwd = 2, bty = "n")

```



Holling IV's curve is the “dramatic spike” referenced later in Act 3 – it's easy to miss on a default y-axis, so it's worth zooming in near $x = 9\text{--}13$ if a student's plot looks flat.

2b. SSR table (5 tank-size means):

```

ssr <- function(pred) sum((obs - pred)^2)
c(
  Ricker          = ssr(ricker(xs, a_rk, b_rk)),
  FlexibleRicker  = ssr(flex_ricker(xs, a_fr, p_fr, c_fr)),
  HollingIV       = ssr(holling4(xs, a_h4, b_h4, c_h4))
)

```

```

##          Ricker FlexibleRicker      HollingIV
##    6.73857019    0.11333450    0.08443968

```

Candidate	SSR
Ricker ($a = 1, b = 0.15$)	6.74
Flexible Ricker	0.11

Candidate	SSR
Holling type IV	0.08

2c. By eye, the flexible Ricker (orange) looks the most reasonable – a smooth hump that tracks the general shape of the data. Holling IV (green) has the *lowest* SSR of the three, but its curve is a narrow, biologically implausible spike. So “looks best” and “lowest SSR” point to *different* candidates here – worth calling out explicitly if a student’s answer conflates the two.

2d. The core idea students should land on: with only 5 data points and 3 free parameters, a curve family has very little left to constrain it (essentially 2 residual degrees of freedom), so it can nearly interpolate the data regardless of whether the resulting shape is a sensible model of the underlying process. Lowest SSR is not the same as “best model” – a model that fits its own training data almost perfectly, especially one with many parameters relative to the data, should raise suspicion rather than confidence. What you’d want to check: how the fit holds up on data it wasn’t fit to (out-of-sample performance), or how stable the fitted parameters are under resampling – which is what Act 3 does next.

Group discussion. No fixed answer – have students reconcile differing SSR values (should match closely if everyone used the given parameter values and the 5-point convention) and discuss their reasoning in 2d.